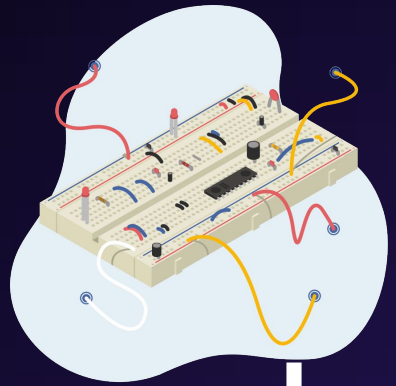
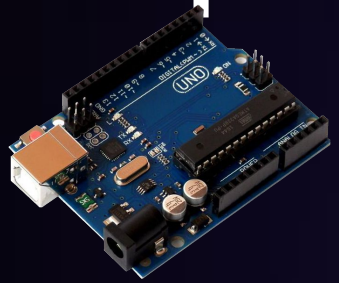
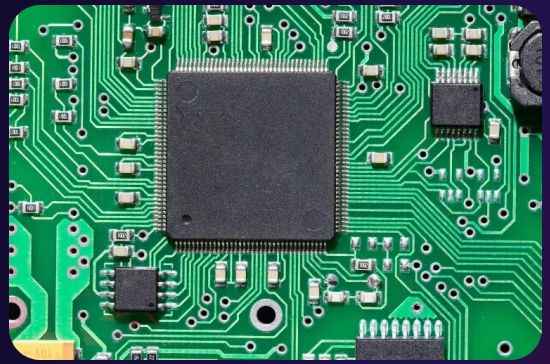
Architecture of electronic and computing machine and computing systems







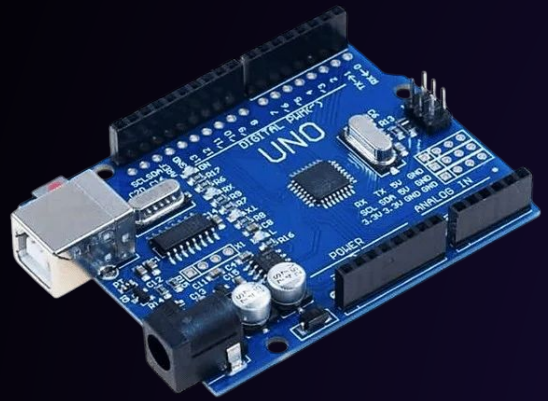
Course Code: S002
Author: Mirlan Nurbekov

ALA-TOO INTERNATIONAL UNIVERSITY
IT & BUSINESS COLLEGE

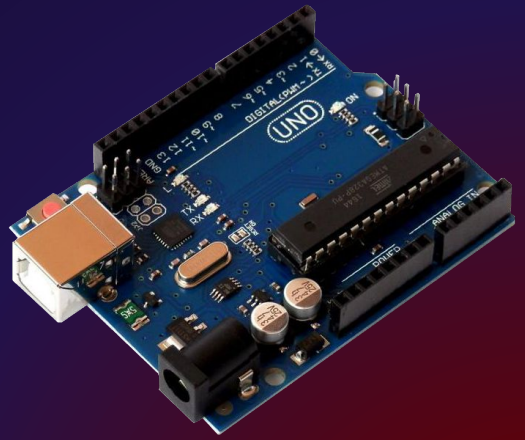
What is a microcontroller (MCU)



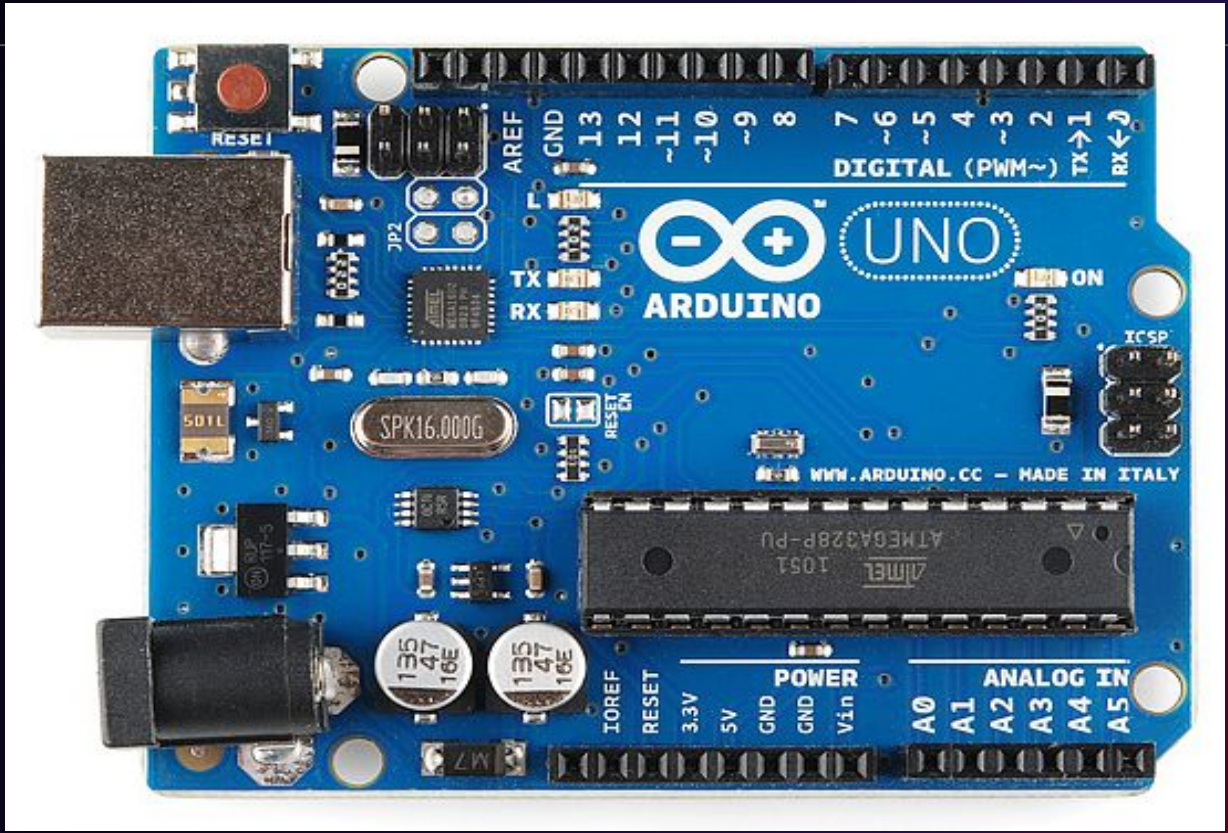
Agenda



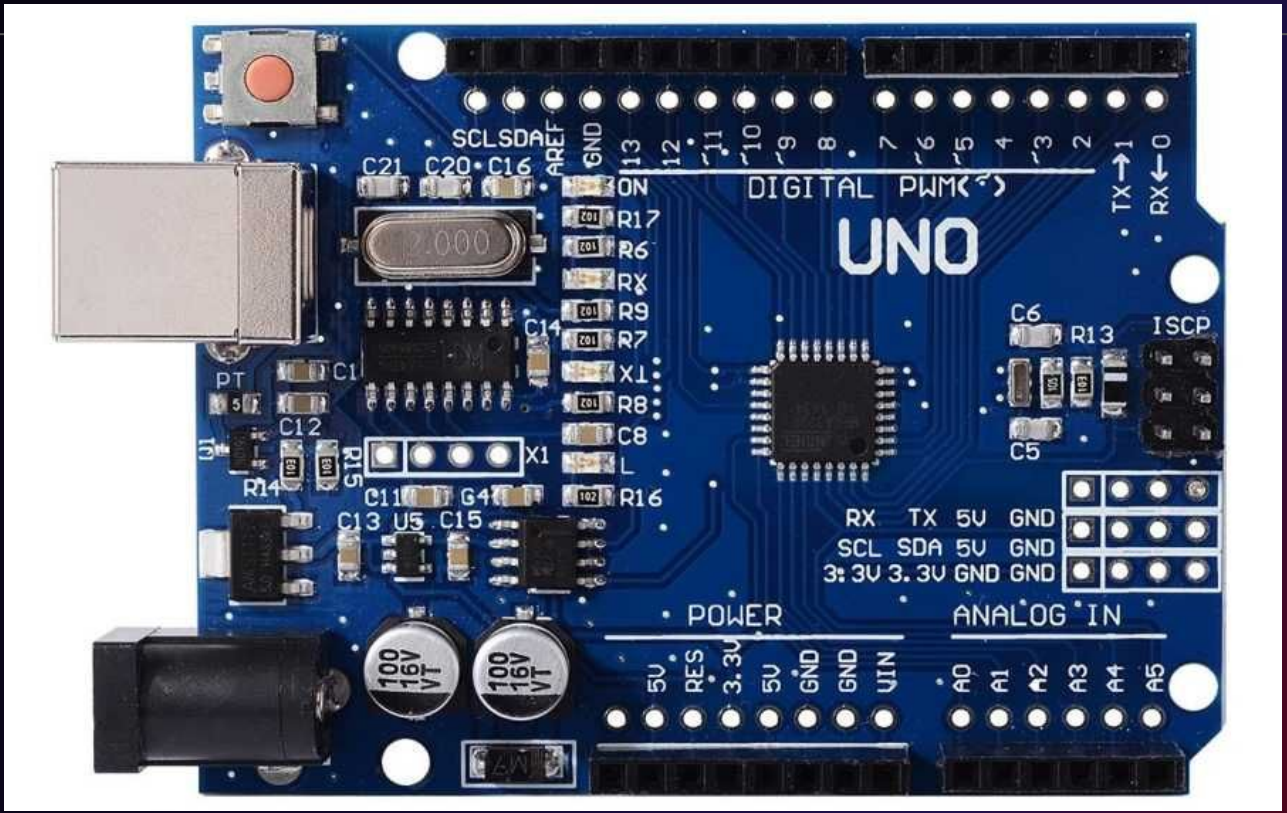
ARDUINO



Arduino UNO


RESET
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
JP2
TX
RX
ARDUINO
UNO
ON
ICSP
SPK16.000G
RESET EN
WWW.ARDUINO.CC – MADE IN ITALY
ATMEGA328P-PU
POWER
ANALOG IN
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

Arduino UNO smd



ATmega328P

8-bit Processor

32Kb Memory



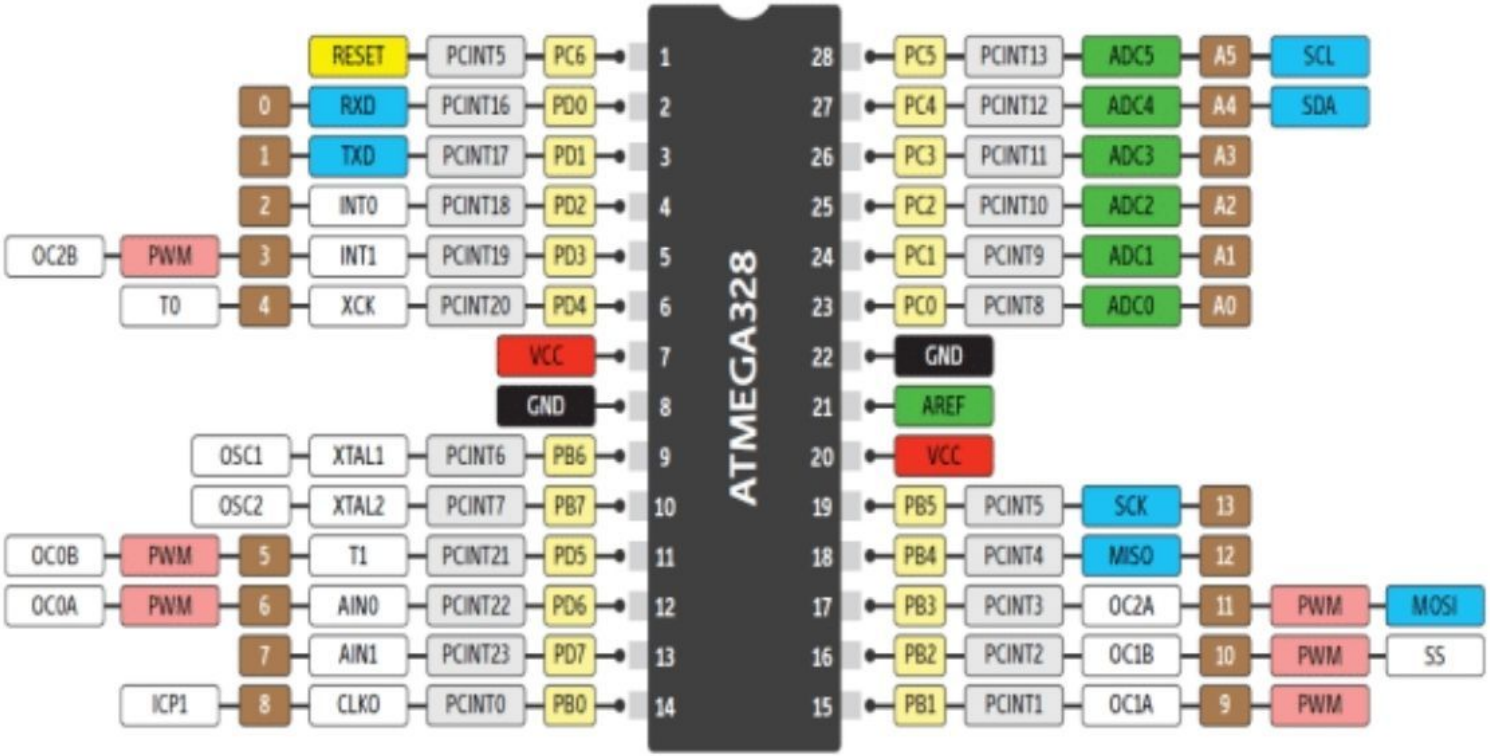
16MHz Clock Speed

Why do we choose Arduino Uno



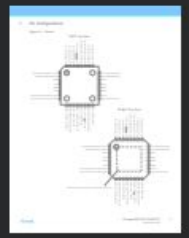
Why do we choose Arduino Uno

Arduino Uno uses the ATmega328P microcontroller because it offers the right balance of performance, low power consumption, and cost. It has enough memory and I/O pins for most beginner and intermediate projects, is widely supported by open-source tools, and is reliable for both learning and prototyping.



Atmega328 Pinout







1.1 Pin Descriptions

1.1.1 VCC

Digital supply voltage.

1.1.2 GND

Ground.

1.1.3 Port B (PB7:0) XTAL1/XTAL2/TOSC1/TOSC2

Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Depending on the clock selection fuse settings, PB6 can be used as input to the inverting oscillator amplifier and input to the internal clock operating circuit.

Depending on the clock selection fuse settings, PB7 can be used as output from the inverting oscillator amplifier.

If the internal calibrated RC oscillator is used as chip clock source, PB7..6 is used as TOSC2..1 input for the asynchronous Timer/Counter2 if the AS2 bit in ASSR is set.

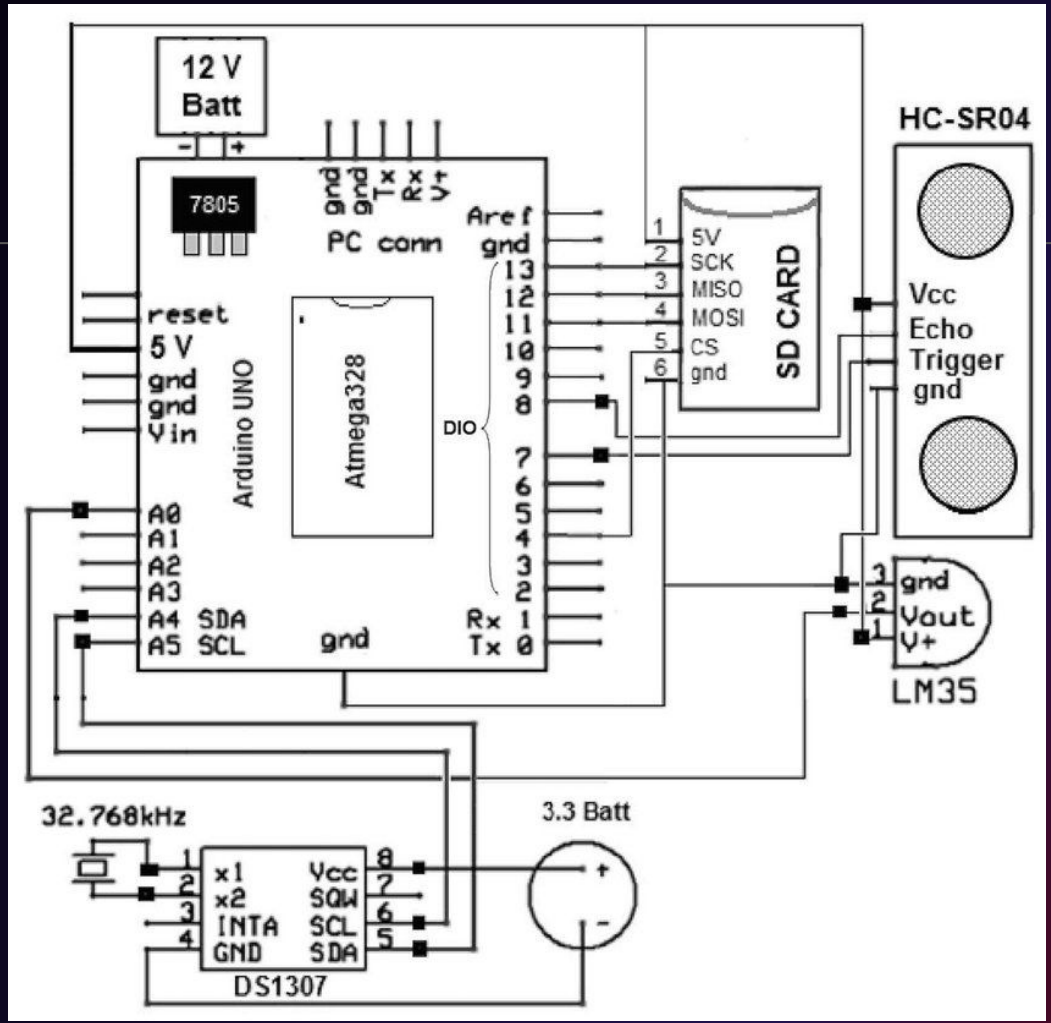
The various special features of port B are elaborated in Section 13.3.1 “Alternate Functions of Port B” on page 65 and Section 8. “System Clock and Clock Options” on page 24.

1.1.4 Port C (PC5:0)

Port C is a 7-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The PC5..0 output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The port C pins are tri-stated when a reset condition becomes active, even if the clock is not running.

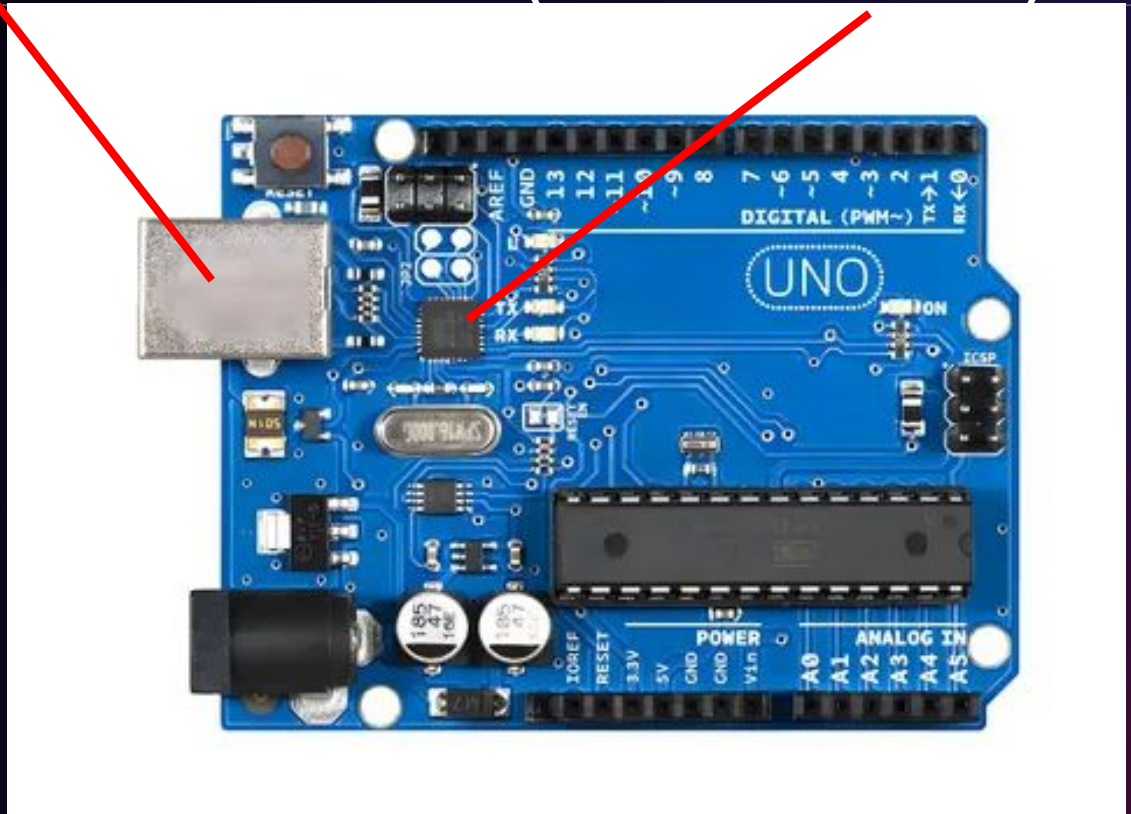
1.1.5 PC6/$\overline{\text{RESET}}$

If the RSTDISBL fuse is programmed, PC6 is used as an input pin. If the RSTDISBL fuse is unprogrammed, PC6 is used as


12 V
Batt
7805
gnd
gnd
Tx
Rx
V+
PC conn
Aref
gnd
13
12
11
10
9
8
7
6
5
4
3
2
Rx 1
Tx 0
reset
5 V
gnd
gnd
Vin
Arduino UNO
Atmega328
DIO
A0
A1
A2
A3
A4 SDA
A5 SCL
gnd
1 5V
2 SCK
3 MISO
4 MOSI
5 CS
6 gnd
SD CARD
HC-SR04
Vcc
Echo
Trigger
gnd
gnd
Vout
V+
LM35
32.768kHz
x1
x2
INTA
GND
Vcc
SQW
SCL
SDA
DS1307
3.3 Batt
+
-

USB PORT

USB-serial converter (UART serial) - TX/RX

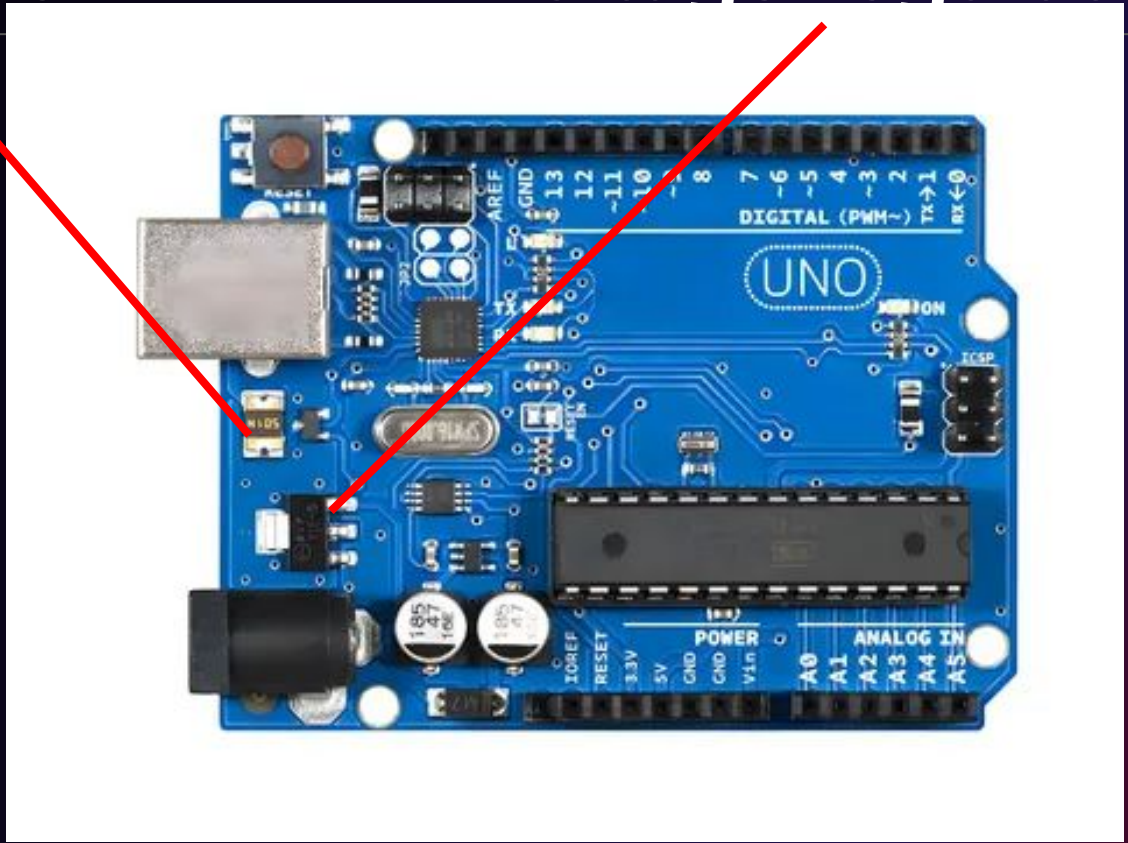


USB port - is used for two main purposes: to program the microcontroller by uploading code from a computer, and to provide power to the board. It also allows serial communication for debugging and data exchange with the PC.

USB-serial converter (UART serial) - TX/RX
The ATmega328P works with UART serial (TX/RX pins), not USB directly. The converter translates between USB signals from the computer and UART signals the microcontroller understands.
UART stands for Universal Asynchronous Receiver-Transmitter. It is a hardware communication protocol that sends and receives data serially using TX (transmit) and RX (receive) lines.

Polyfuse

Voltage regulator



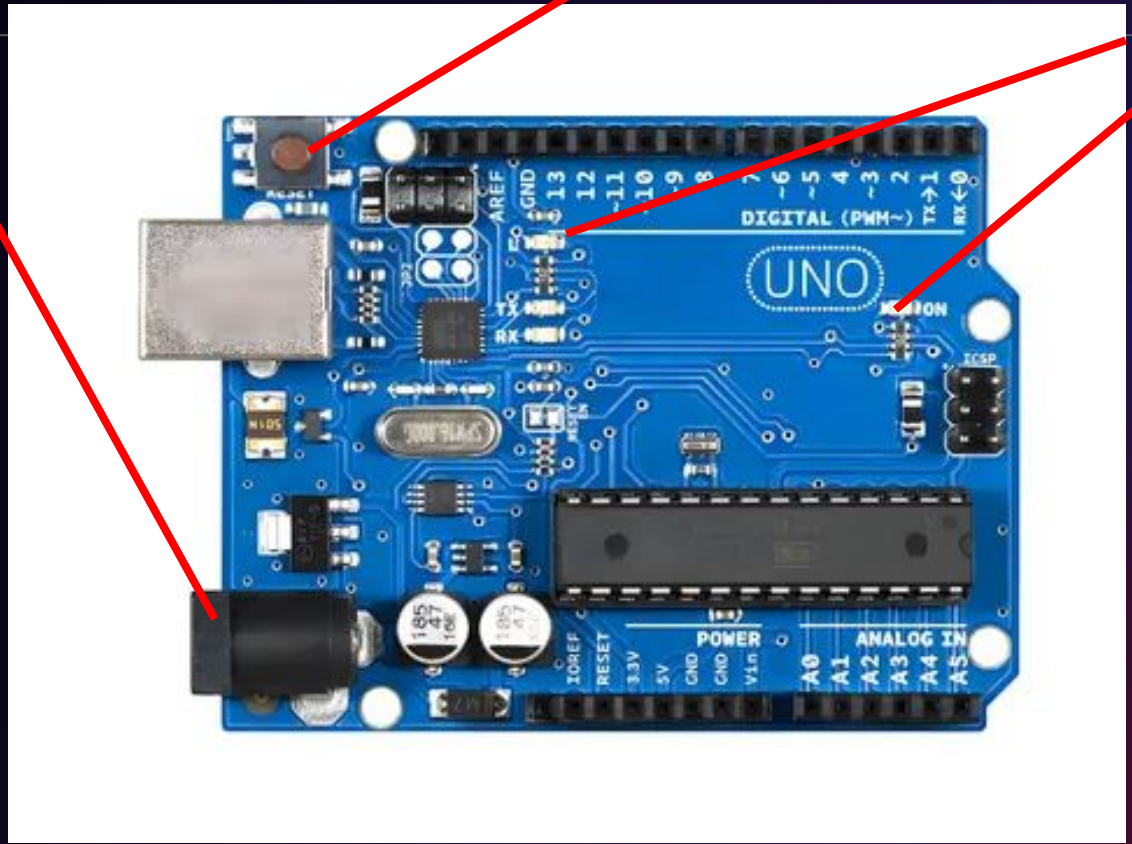
A **polyfuse** is a resettable fuse that protects the Arduino from overcurrent. If too much current flows, it temporarily disconnects power, then resets itself when the fault is cleared.

The **voltage regulator** ensures the Arduino always receives a stable 5V supply. It protects the board by converting higher input voltages (like 7-12V) down to safe operating levels.

DC Barrel Jack

RESET BTN

LED

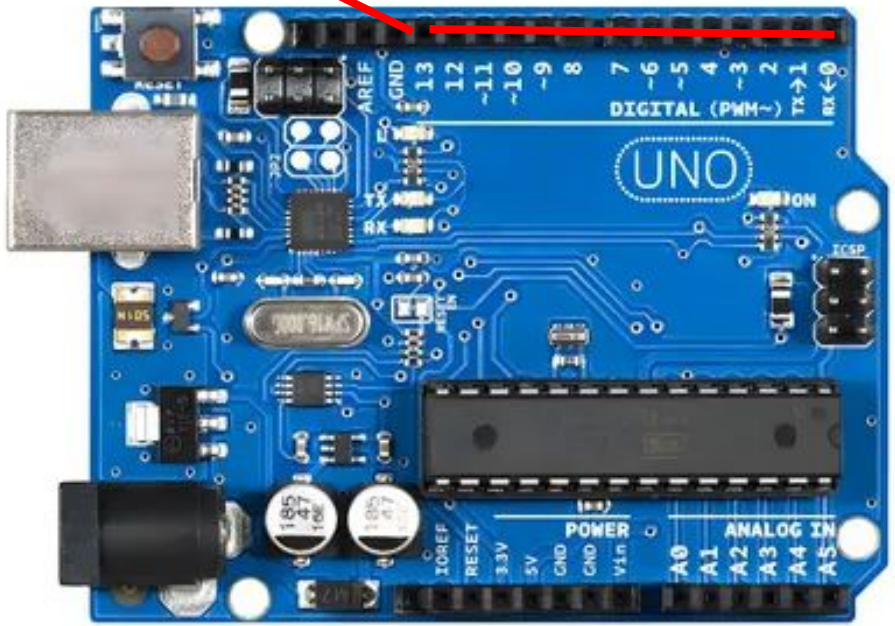


The DC barrel jack allows the Arduino to be powered from an external adapter (7-12V). The onboard voltage regulator then converts this input to a safe 5V for the board.

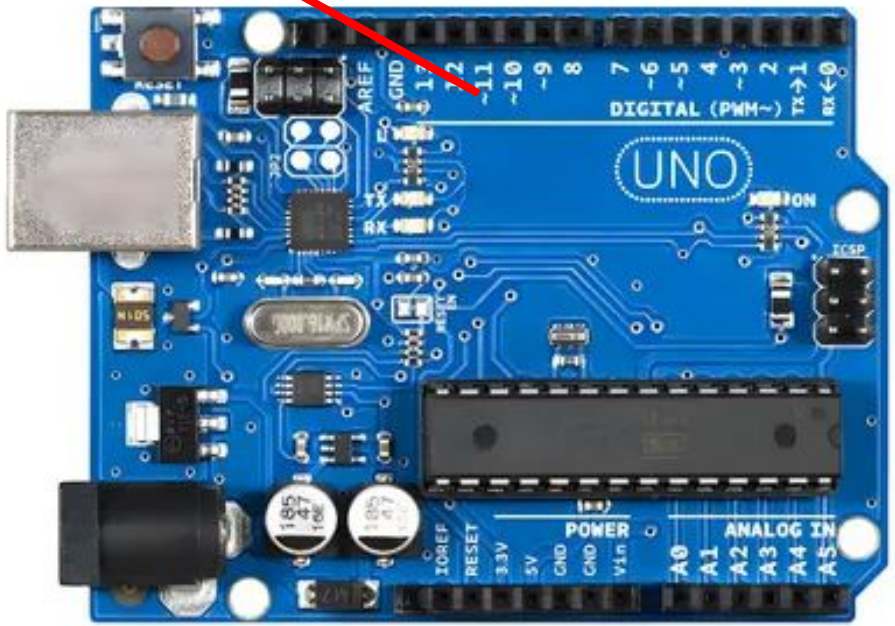
The **RESET button** restarts the Arduino by resetting the microcontroller. It stops the current program and runs the uploaded code again from the beginning.

The **built-in LED** (usually on pin 13) is used as a simple indicator. It helps test code, debug programs, and confirm that the Arduino is running.

DIGITAL I/O PINS



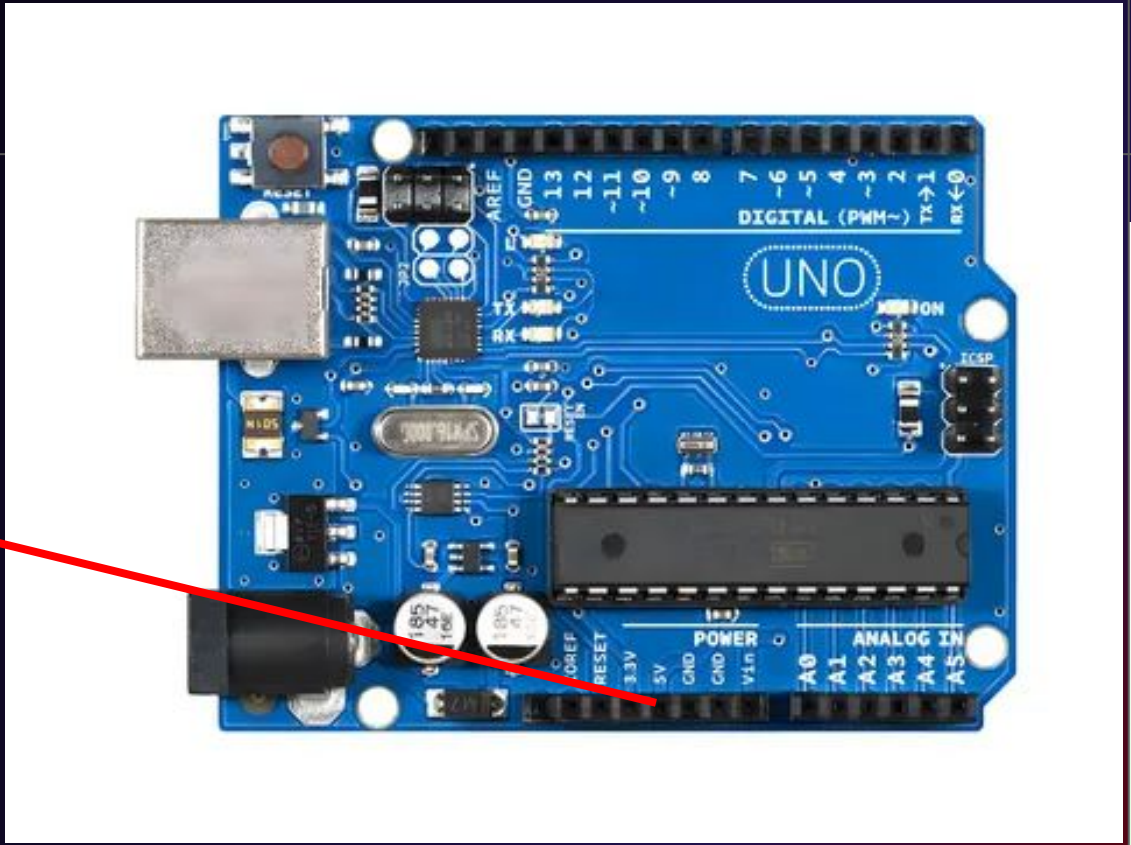
~Tilde



Digital **I/O pins** can be configured as either input or output. They read digital signals (HIGH/LOW) from sensors or switches, and control devices like LEDs, motors, and relays.

Pins marked with ‘~’ support PWM (Pulse Width Modulation). PWM lets the Arduino simulate analog output, for example to dim LEDs or control motor speed.

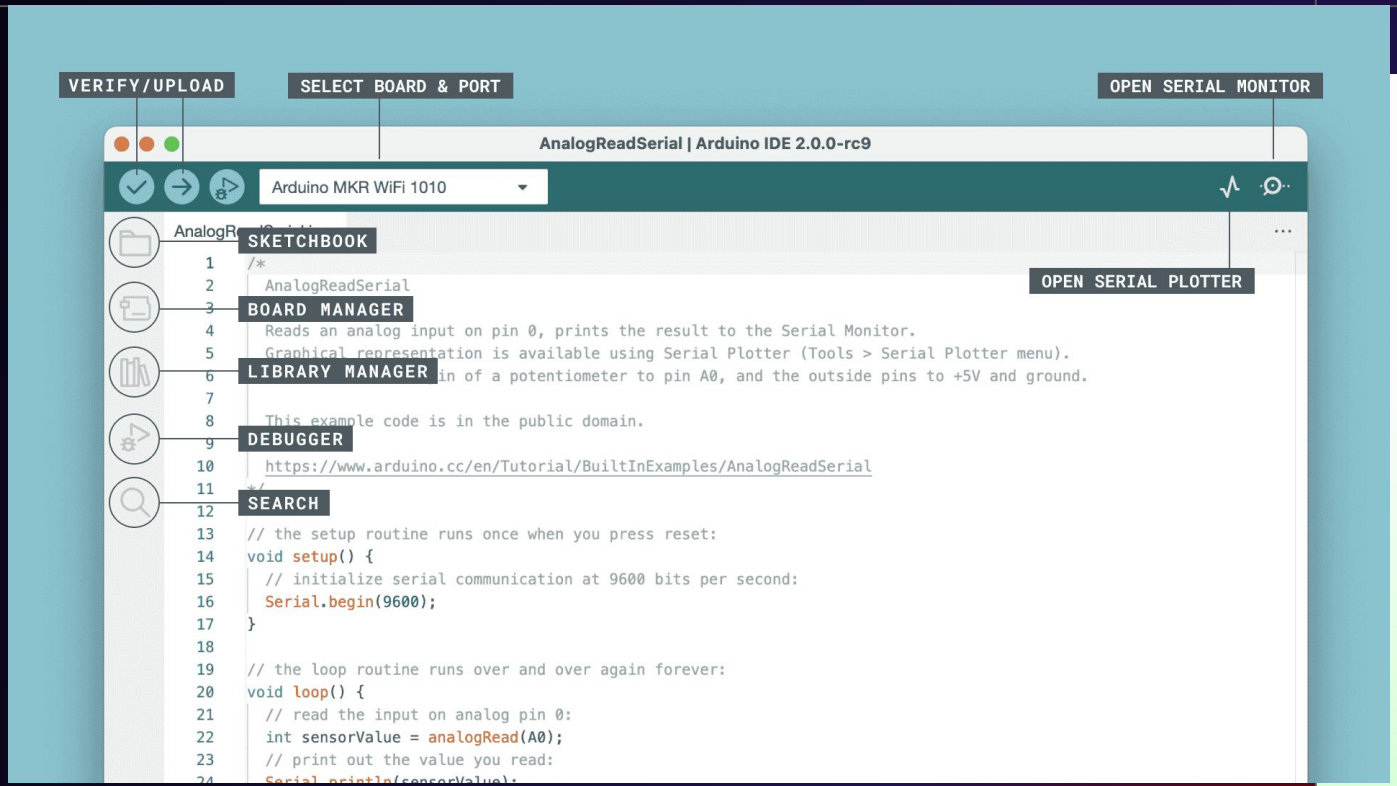
Power



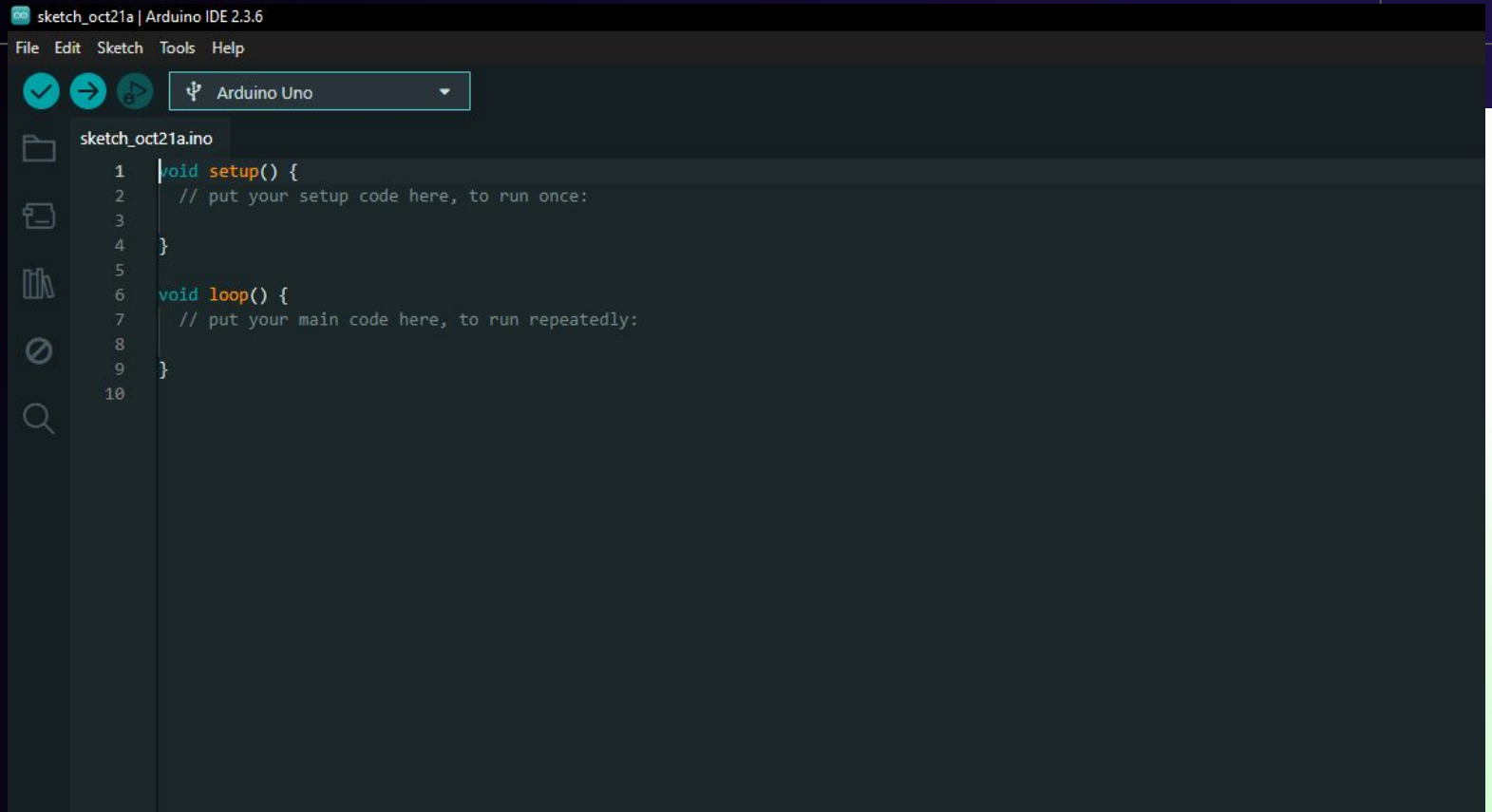
Arduino IDE
(Integrated Development Environment)



Arduino IDE


VERIFY/UPLOAD
SELECT BOARD & PORT
OPEN SERIAL MONITOR
AnalogReadSerial | Arduino IDE 2.0.0-rc9
Arduino MKR WiFi 1010
SKETCHBOOK
BOARD MANAGER
LIBRARY MANAGER
DEBUGGER
SEARCH
OPEN SERIAL PLOTTER
/*
AnalogReadSerial
Reads an analog input on pin 0, prints the result to the Serial Monitor.
Graphical representation is available using Serial Plotter (Tools > Serial Plotter menu).
in of a potentiometer to pin A0, and the outside pins to +5V and ground.
This example code is in the public domain.
https://www.arduino.cc/en/Tutorial/BuiltInExamples/AnalogReadSerial
*/
// the setup routine runs once when you press reset:
void setup() {
// initialize serial communication at 9600 bits per second:
Serial.begin(9600);
}
// the loop routine runs over and over again forever:
void loop() {
// read the input on analog pin 0:
int sensorValue = analogRead(A0);
// print out the value you read:

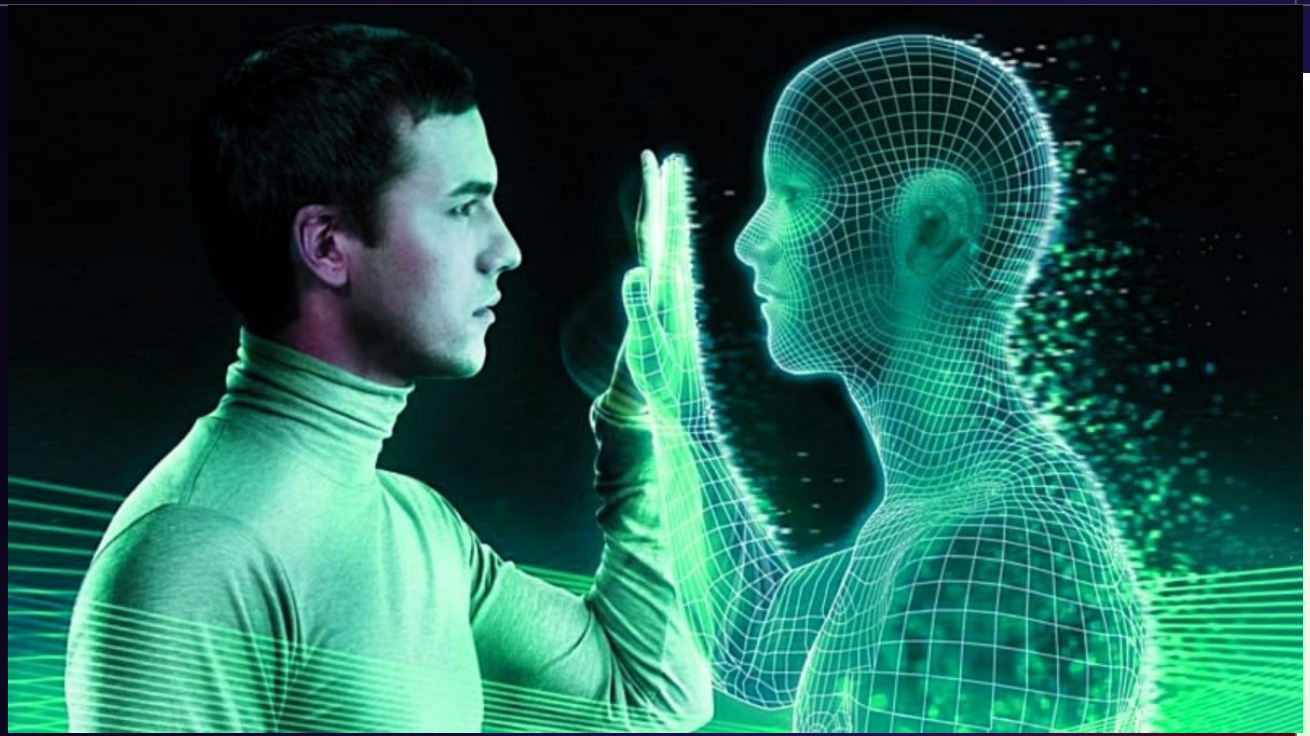
Arduino IDE


sketch_oct21a | Arduino IDE 2.3.6
File Edit Sketch Tools Help
Arduino Uno
sketch_oct21a.ino
1 void setup() {
2 // put your setup code here, to run once:
3
4 }
5
6 void loop() {
7 // put your main code here, to run repeatedly:
8
9 }
10

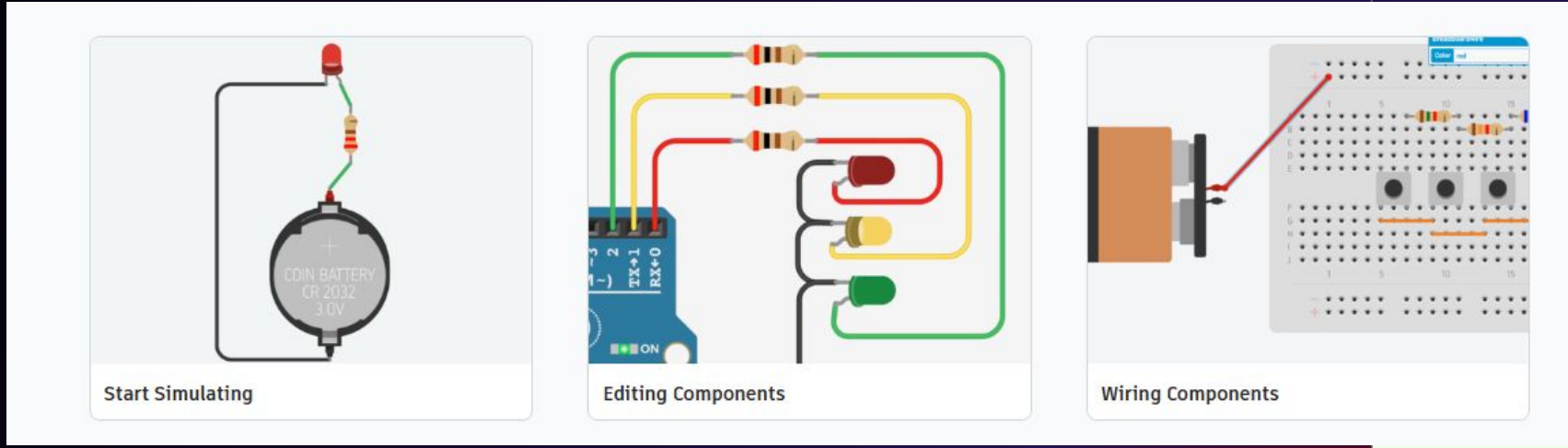
Arduino IDE

```
1  void setup() {
2    // put your setup code here, to run once:
3
4  }
5
6  void loop() {
7    // put your main code here, to run repeatedly:
8
9  }
```

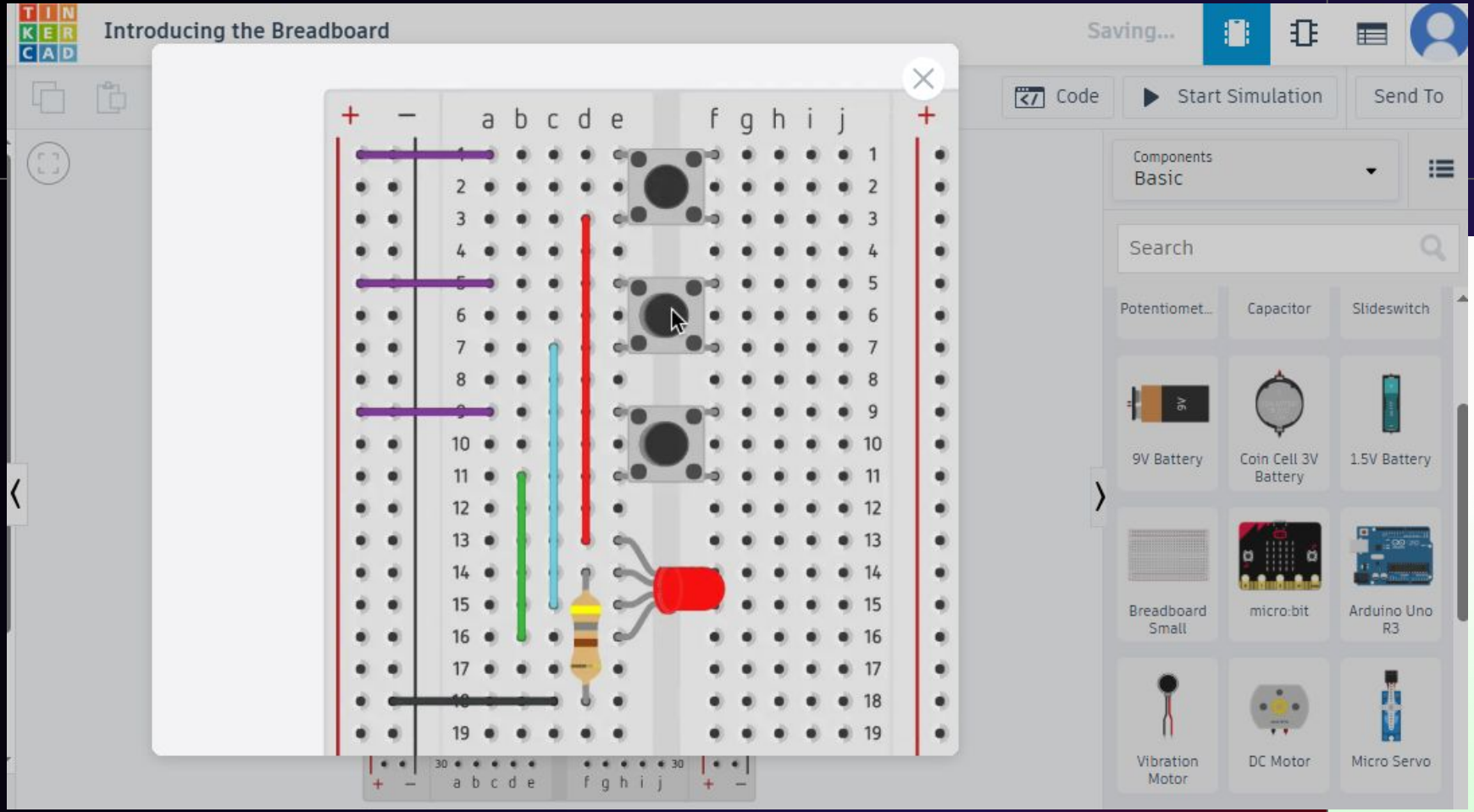
Simulation options



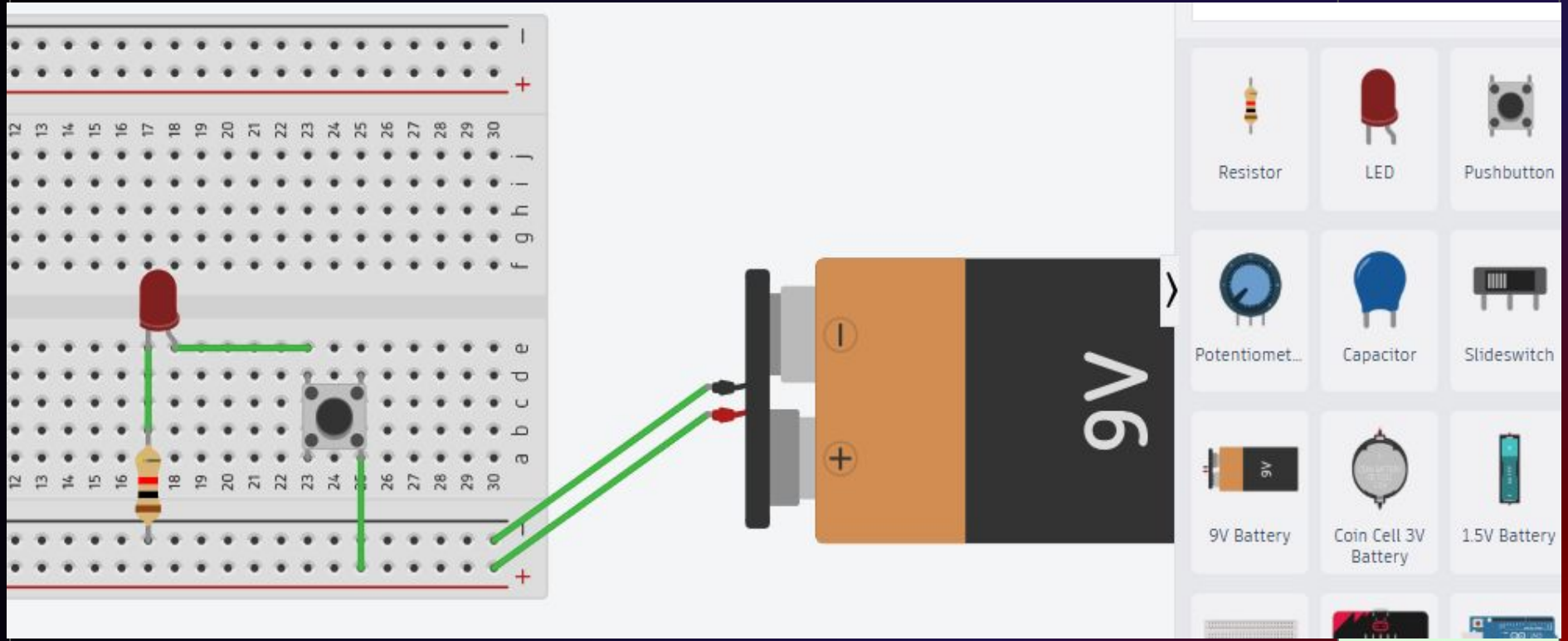
AUTODESK Tinkercad



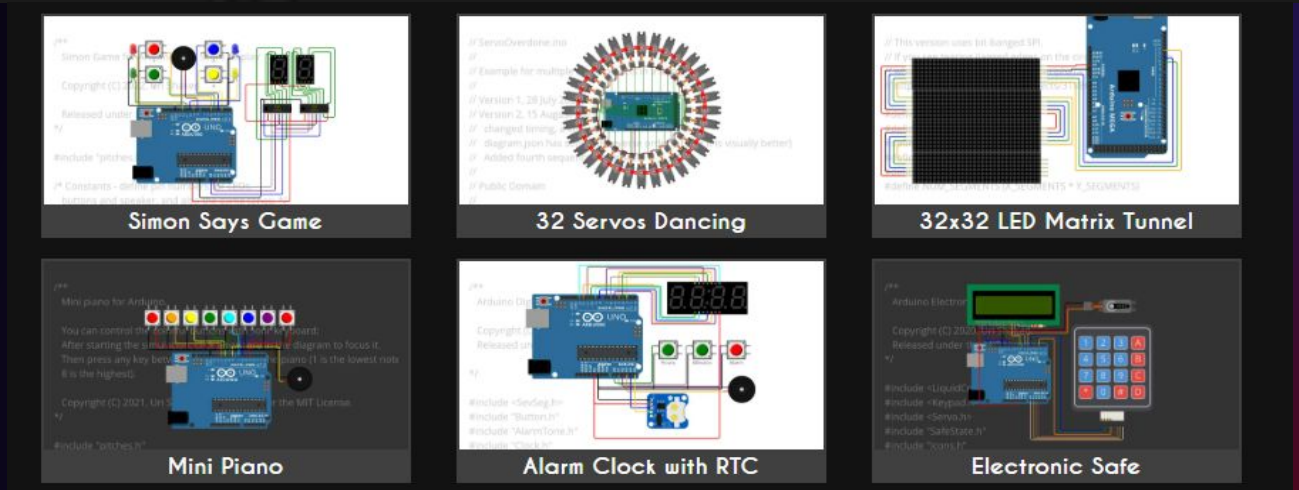
https://www.tinkercad.com/circuits


Introducing the Breadboard
Saving...
Code
Start Simulation
Send To
Components
Basic
Search
Potentiomet...
Capacitor
Slideswitch
9V Battery
Coin Cell 3V Battery
1.5V Battery
Breadboard Small
micro:bit
Arduino Uno R3
Vibration Motor
DC Motor
Micro Servo

Simulations using TinkerCad



WOKWI



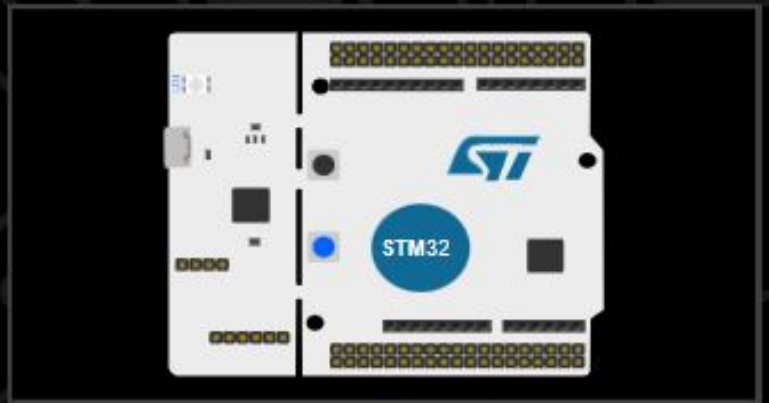
https://wokwi.com/



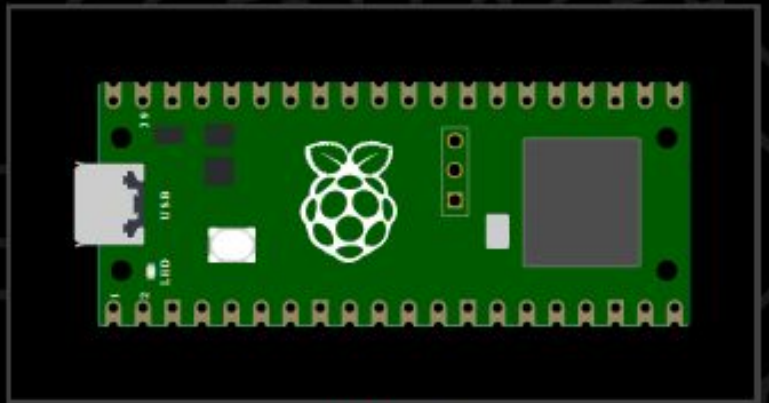
Arduino (Uno, Mega, Nano)



ESP32

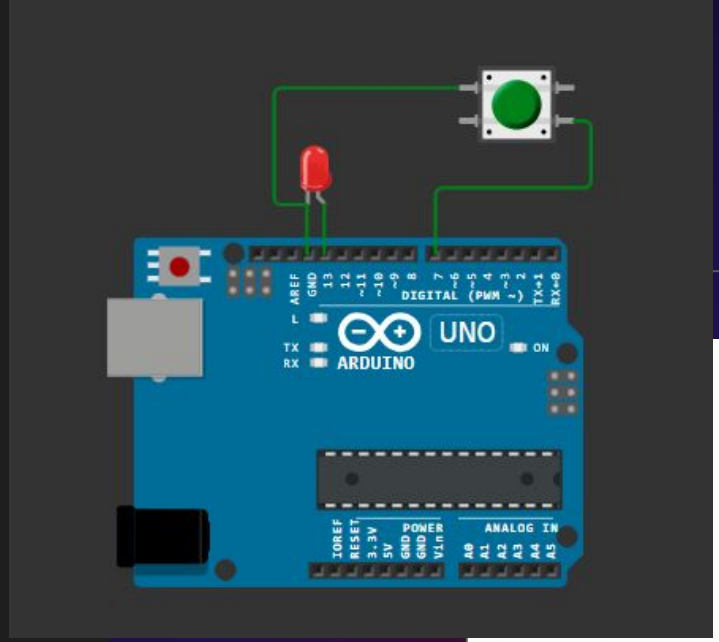


STM32



Pi Pico

```
1
2   int led = 13;
3   int btn = 7;
4   static int led_state, prev_btn_data;
5
6   void setup() {
7     pinMode(led, OUTPUT);
8     pinMode(btn, INPUT_PULLUP);
9   }
10
11  void loop() {
12    int btn_data = digitalRead(btn);
13    if(btn_data == LOW && btn_data!=prev_btn_data)
14    {
15      led_state = !led_state;
16    }
17    digitalWrite(led, led_state);
18    prev_btn_data = btn_data;
19  }
```



Self test mcq

1. Which microcontroller is used by the Arduino Uno (original)?

   A) ATmega328P B) ESP32 C) STM32 D) ATmega2560

2. Which two lines are used for UART serial communication?

   A) TX/RX B) SDA/SCL C) MOSI/MISO D) VCC/GND

Self test mcq

3. What is the main purpose of the Arduino's polyfuse?

   A) Store program data

   B) Protect from overcurrent

   C) Regulate clock speed

   D) Network control PWM output

4. What does the ~ symbol on Arduino UNO pins indicate?

   A) UART communication

   B) PWM support

   C) Analog input

   D) Power output

Self test mcq

5. What happens when the RESET button is pressed?

A) USB is disconnected

B) The MCU restarts

C) The voltage increases

D) The LED is removed

Answers

1. A
2. A
3. B
4. B
5. B

Home work

Read the Arduino UNO and ATmega328P documentation/datasheet and answer following questions:

1 - Identify the ATmega328P and its key specifications.

2 - Explain what 8-bit means in the context of the processor.

3 - Explain the purpose of Flash memory.

4 - Explain TX and RX in UART communication.

5 - Explain what happens when RESET is pressed.

Home work

Read the Arduino UNO and ATmega328P documentation/datasheet and answer following questions:

6 - Explain the purpose of the ~ pins and give an example.

7 - Explain the role of the voltage regulator.

8 - If I want to control LED brightness. Which Arduino pin capability should they use, and why?